$$\lambda' = \lambda - v_S T = \lambda - \left(\frac{v_S}{f}\right)$$

The observed frequency is then given by

$$f' = \frac{v}{\lambda'} = \frac{v}{\lambda - \left(\frac{v_S}{f}\right)}$$

$$= \frac{v}{\left(\frac{v}{f}\right) - \left(\frac{v_S}{f}\right)}$$

$$f' = \left(\frac{v}{v - v_S}\right) f \qquad (11.86)$$

(source moving toward the listener)

Thus, whenever the source moves toward the stationary listener, the observed frequency is greater than the source frequency.

If the source is moving away from the stationary listener, the observed frequency can be obtained from equation (11.86) by taking negative value for $v_S$. It is given by

$$f' = \left(\frac{v}{v - (-v_S)}\right) f$$

$$f' = \left(\frac{v}{v + v_S}\right) f \qquad (11.87)$$

(source moving away from the listener)

Thus, the observed frequency is less than the source frequency when the source is moving away from the stationary listener.

### iii) Observed frequency: Both source and listener moving

When both source and listener are moving, the observed frequency is obtained by combining equations (11.84) and (11.86).

$$f' = \left(\frac{v + v_L}{v - v_S}\right) f \qquad (11.88)$$

The sign convention we used here is that both $v_S$ and $v_L$ take positive values if the source or the listener moves toward the other. Likewise, they are negative when the source or the listener moves away from the other.

The observed frequency for different situations of relative motion between the source and the listener is consolidated in Table 11.4.




It is important to note that the change in frequency occurs either due to the change in speed of sound (when the listener moves and source at rest) or due to the change in wavelength of sound (when the source moves and observer at rest).

If both source and listener move, the change in frequency occurs due to both the change in speed of sound and the change in wavelength of sound wave.




Suppose the source moves faster than sound (that is, the source is supersonic), the equations (11.84) and (11.86) for observed frequency will become invalid and a stationary listener in front of the source hears no sound as the sound waves are at the rear of the source.

At such speeds, the newly produced waves and the old waves interfere constructively which leads to very large amplitude of sound, called a 'sonic boom' or 'shock wave'.



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